Jaya Sai Krishna Muppalla

Staff Software Engineer

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SUMMARY

Staff Software Engineer with 11 years designing and operating distributed systems and cloud infrastructure for Samsung Electronics at consumer scale. Builds Kubernetes platforms, cloud-native backend services, streaming data pipelines, and applied AI/RAG systems, and owns the CI/CD, observability, and on-call practices behind them.

SKILLS

Languages & Frameworks: Java, Node.js, TypeScript, Go, Python, Spring Boot, Express, NestJS, Spring WebFlux

Cloud & Infrastructure: AWS, Azure, OCI, Kubernetes, Helm, Docker, KubeVirt, Terraform, Serverless Framework, Packer

Data & Messaging: Kafka, Kinesis, Apache Flink, MongoDB, Redis, MySQL, Elasticsearch

Observability & CI/CD: OpenTelemetry, CloudWatch, PagerDuty, Prometheus, Grafana, Traefik, GitLab CI/CD, GitHub Actions, Jenkins

APIs & Protocols: REST, GraphQL, WebSocket, gRPC, MQTT, WebRTC, RTSP, SCTE-35, VAST

AI & Data Retrieval: Agentic AI, Claude, Amazon Bedrock, Codex, LangChain, Azure OpenAI, OpenAI API, Retrieval-Augmented Generation (RAG), Vector Search

PROFESSIONAL EXPERIENCE

Samsung Electronics America

July 2019 – Present

Staff Engineer, Back-End Server

Feb 2025 – Present

- Architected a server-side, ad-supported linear "FAST channel" streaming platform for Samsung Smart TVs on Node.js/NestJS and Kubernetes (EKS), assembling looping and scheduled playlists into continuous live streams delivered via HLS through a CDN-backed origin.
- Implemented server-side ad insertion (SSAI) using SCTE-35 markers and built a scheduling engine for fixed-time-window content blocks with automatic filler coverage, enabling live, in-place channel editing with zero stream interruption.
- Built a custom Kubernetes controller that reconciles published channels in MySQL with corresponding Kubernetes CRDs, and a VAST-compliant ad service routing TV ad inventory to a portfolio of interactive mini-games served as playable ad creative.
- Designed a real-time analytics pipeline (AWS Kinesis, S3, Glue, Athena) capturing VAST tracking events, with QuickSight dashboards for playable-ad performance reporting.
- Developed a GPT-based movie recommendation system using custom Retrieval-Augmented Generation (RAG) with LangChain, Azure OpenAI, and vector search (Azure AI Search, Redis).
- Owned a seven-version, zero-downtime Kubernetes upgrade (1.26 to 1.33) with a staged rollout, per-step validation, and a documented rollback runbook.
- Manage infrastructure as code with Terraform and Helm as the single path to production.
- Carry on-call for these backend systems, shaping alerting from real incidents and automating recurring remediations to cut repeat pages.
- Migrated observability to OpenTelemetry for unified logs, metrics, and traces, with CloudWatch dashboards and PagerDuty integration for on-call routing.

Engineer 3, Software Process Engineer March 2022 – Jan 2025

- Engineered automation for virtual game machine provisioning using KubeVirt (Kubernetes + VMs) across AWS and OCI bare-metal, running games on Nvidia and AMD GPUs via vGPU/SR-IOV on Windows and Linux.
- Automated optimized EKS/OKE machine images for game streaming using Packer (qcow2 format), including host GPU driver management, to streamline deployment.
- Designed a custom Kubernetes host-port operator to make game machines accessible economically, and containerized Windows games to run on Linux using Wine.
- Established and managed a Rook + Ceph storage cluster to support Read-Only-Many (ROX) persistent volumes, compensating for a gap in native AWS support.
- Extended the Puppetlabs PVPool controller with Kubernetes volume snapshots, enabling near-instant PVC replication for efficient game data hosting.
- Delivered sub-100ms end-to-end latency for interactive cloud gameplay across the platform.

Engineer 2, Software Process Engineer July 2019 – Feb 2022

- Built and operated Node.js backend services on Kubernetes (EKS), deployed via Helm, powering B2B applications across Samsung's wearable platform.
- Built an indoor positioning system using beacon technology for proximity detection and contributed to contact-tracing solutions for wearable devices.
- Used Apache Flink for real-time stream processing with Complex Event Processing (CEP) for advanced event pattern detection.
- Designed event-driven microservices on a Function-as-a-Service (FaaS) model on AWS, with deployment automation via the Serverless framework.
- Built fully reactive applications with Spring WebFlux, MongoDB, and Kafka, and designed GraphQL APIs for client data access.
- Engineered multi-region traffic infrastructure with Terraform to meet regulatory compliance requirements, and ran the team's private npm/Helm artifact repositories (Sinopia, S3).

First Object Inc **Aug 2015 – July 2019**

Server Engineer (Contracted to Samsung Electronics America)

- Developed and deployed a serverless stack across multiple global regions to minimize latency, with GeoSpatial queries and GeoHashing powering geofence and location-based alerting.
- Integrated mapping APIs including OpenStreetMap, Leaflet, Google Maps, and HERE Maps for geocoding and data visualization.
- Built Spring Boot microservices and IoT applications using the MQTT protocol for device communication.
- Designed a reactive, real-time UI using Meteor, Handlebars, and MongoDB, and contributed across the stack including CMS (Magnolia) and Kafka streaming.
- Conducted rapid prototyping of proof-of-concept systems and matured them into production-grade AWS deployments.

SELECTED PROJECTS

FAST Channel Streaming Platform · **Playable Ads & Mini-Games Platform** · **GPT Movie Recommendation Engine** · **Cloud Game Streaming Infrastructure** · **Reactive Wearables & IoT Backend** · **Global Serverless Geospatial Backend**

EDUCATION

Master of Science, Computer Science, University of Missouri Kansas City (UMKC) **Aug 2013 – May 2015**

Bachelor of Technology, Electronics and Communication Engineering, Jawaharlal Nehru Technological University **May 2009 – April 2013**